

EGGROLL Convergence

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1 Problem Setup

Sarkar et al. (2025) introduce EGGROLL as a low-rank evolutionary strategies method for scalable black-box optimization of large neural networks, with major applications being LLM fine tuning and post-training. In this excerpt, we prove that (symmetric) EGGROLL converges to a first order stationary point.

Assumption 1 (Properties of f) *We suppose that $f : \mathbb{R}^{d \times m} \rightarrow \mathbb{R}$ satisfies the following properties:*

1. f is differentiable and lower bounded, i.e. $f^* = \min_x f(x) > -\infty$
2. f is L -gradient Lipschitz, i.e.

$$\|\nabla f(X) - \nabla f(Y)\|_F \leq L\|X - Y\|_F \quad \forall X, Y \in \mathbb{R}^{d \times m}$$

Definition 1 *A point $X \in \mathbb{R}^{d \times m}$ is an ε -first order stationary point if*

$$\|\nabla f(X)\|_F \leq \varepsilon$$

In this excerpt we work with symmetric EGGROLL, following the low-rank perturbation framework of Sarkar et al. (2025).

Definition 2 (Symmetric EGGROLL Update) *We define a rank r symmetric zeroth order estimator as follows:*

$$g_\sigma^r(X) := \frac{1}{r} \sum_{i=1}^r \frac{f(X + \sigma P_i) - f(X - \sigma P_i)}{2\sigma} P_i$$

using $P_i = a_i b_i^\top$ where $a_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ and $b_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_m)$, and $\sigma > 0$ is the smoothing radius.

Algorithm 1 Rank- r Symmetric EGGROLL

Require: Initial point $X_0 \in \mathbb{R}^{d \times m}$, horizon T , stepsize $\eta > 0$, smoothing radius $\sigma > 0$, rank $r \in \mathbb{N}$
for $t = 0, \dots, T - 1$ **do**
 Sample $\{a_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ and $\{b_{t,i}\}_{i=1}^r \sim \mathcal{N}(0, I)$ to compute $g_\sigma^r(X_t)$
 Update $X_{t+1} \leftarrow X_t - \eta g_\sigma^r(X_t)$
end for
return X_T

We state an informal version of the main result.

Theorem 3 *Consider running Algorithm 1. Suppose $0 < \delta \leq 1/e$. Suppose*

$$\sigma = \tilde{O}\left(\frac{\varepsilon}{dm}\right), \quad \eta = \tilde{O}\left(\frac{r}{dm}\right).$$

Then in

$$T = \tilde{\Omega}\left(\frac{dmL(f(X_0) - f^*)}{r\varepsilon^2}\right)$$

iterations (with each iteration using $2r$ function evaluations), with probability at least $1 - \delta$, at least $3/4$ of the iterates are ε -first order stationary points.

2 Proof Outline

First, we prove a few concentration bounds which will aid us in bounding the error terms. The lion's share of work went into proving Theorem 6, which is the main tool used to get the running time above. We then introduce a preliminary lemma on function decrease, Lemma 11. Because the update could either decrease function value if it aligns sufficiently with the gradient, or it could increase it but by an amount of the same order as other error terms, we use Lemma 15 and Lemma 17 to handle these two cases. Armed with these lemmata, Lemma 18 is the remaining work to be done, and we arrive at the formal version of our main theorem, Theorem 19 after a careful choice of parameters.

3 Concentration Bounds

Here we prove a novel concentration bound about the EGGROLL error. Throughout this excerpt, all norms of matrices is the Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} A_{i,j}^2\right)^{1/2}.$$

3.1 Novel Concentration Bound

Throughout this paper, we denote

$$\mathbb{E}_{\mathcal{F}_{t-1}}[X; A] := \mathbb{E}[X\mathbb{I}_A \mid \mathcal{F}_{t-1}]$$

We will need the following lemma from Jin et al. (2019).

Lemma 4 *Let random vectors $x_1, \dots, x_n \in \mathbb{R}^d$, and corresponding filtrations $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ satisfy that*

$$\mathbb{E}_{\mathcal{F}_{t-1}}[x_t] = 0, \quad \mathbb{P}_{\mathcal{F}_{t-1}}(\|x_t\| \geq s) \leq 2e^{-\frac{s^2}{2\sigma^2}} \quad \forall s \in \mathbb{R}$$

Then there exists an absolute constant c such for any fixed $\delta > 0, \theta > 0$, with probability at least $1 - \delta$,

$$\left\|\sum_{i=1}^n X_i\right\| \leq c \cdot \theta \sum_{i=1}^n \sigma_i^2 + \frac{1}{\theta} \log\left(\frac{2d}{\delta}\right).$$

This will help us later.

Lemma 5 *Let X follow χ_d^2 distribution and $0 < \delta < 1/e$. Then there exists L_δ such that for every d , with*

$$A := \{dL_\delta^2 \leq X \leq 3d \log(C/\delta)\}$$

we have

$$\mathbb{E}[(X - d); A] = 0, \quad \mathbb{P}(A) \geq 1 - \delta$$

for some absolute constant C .

Proof Let $f_d(x)$ be the χ_d^2 density,

$$f_d(x) = \frac{1}{2^{d/2} \Gamma(d/2)} x^{d/2-1} e^{-x/2}, \quad x \geq 0.$$

A simple calculation shows that

$$(x - d)f_d(x) = -2 \frac{d}{dx}(xf_d(x))$$

Define $U = 3d \log(1/\delta)$ and define the set

$$A = \{L \leq X \leq U\}$$

We want

$$\mathbb{E}[(X - d); A] = \int_L^U (x - d)f_d(x)dx = -2Lf_d(L) - 2Uf_d(U) \stackrel{\text{want}}{=} 0.$$

Therefore, we need

$$\begin{aligned} Lf_d(L) &= Uf_d(U) \\ \iff L^d e^{-L} &= U^d e^{-U} \\ \iff U - L &= d \log(U/L) \end{aligned}$$

Now define $\ell = L/d$ and $u = U/d$. Then we want to solve

$$\begin{aligned} du - d\ell &= d \log(u/\ell) \\ \iff \ell e^{-\ell} &= u e^{-u} \end{aligned}$$

The function $g(x) = xe^{-x}$ is increasing on $[0, 1)$ and decreasing on $(1, \infty)$. Therefore, as long as $u > 1$, which follows as $\delta < 1/e$, there exists a unique $\ell < 1$ satisfying the above equation. Now notice, as $e^{-\ell} \geq e^{-1}$,

$$\ell e^{-1} \leq \ell e^{-\ell} = 3 \log(1/\delta) \delta^3 \leq 3e^{-1} \delta^2$$

Now we have that

$$\mathbb{P}(A^c) = \mathbb{P}(X \geq U) + \mathbb{P}(X \leq L)$$

Recall the Chernoff bound For $0 < \alpha < 1$,

$$\mathbb{P}(X \leq \alpha d) \leq (\alpha e^{1-\alpha})^{d/2} \leq (e\alpha)^{d/2}$$

Then we have

$$\mathbb{P}(X \leq L) \leq \mathbb{P}(X \leq 3d\delta^2) \leq (3e\delta^2)^{d/2} = (\sqrt{3e}\delta)^d$$

When $\sqrt{3e}\delta < 1$, this is $< \sqrt{3e}\delta$, otherwise this bound holds vacuously. Therefore $\mathbb{P}(X \leq L) \leq \sqrt{3e}\delta$. On the other hand, using that for $t < 1/2$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = (1 - 2t)^{-d/2},$$

along with Markov's inequality, we get

$$\mathbb{P}(X \geq 3d \log(1/\delta)) = \mathbb{P}(\exp(X/3) \geq \exp(d \log(1/\delta))) \leq 3^{d/2} \delta^d \leq \sqrt{3}\delta$$

And thus $\mathbb{P}(A) \geq 1 - C\delta$ for the absolute constant $C = \sqrt{3} + \sqrt{3e}$. ■

Here is the novel concentration bound. The proof is inspired by Proposition 2 of Ren et al. (2023) but with many important differences.

Theorem 6 *Let $\{\mathcal{F}_t\}_{t \geq -1}$ be a filtration, and let $\{a_t\}_{t \geq 0}, \{b_t\}_{t \geq 0}$ be two sequences of independent random vectors such that $a_t \sim \mathcal{N}(0, I_d)$, $b_t \sim \mathcal{N}(0, I_m)$, $a_t, b_t \in \mathcal{F}_t$ and a_t, b_t are independent of $\mathcal{F}_{t-1}$. Let $\{V_t\}_{t \geq 0}$ be a sequence of random matrices such that $V_t \in \mathcal{F}_{t-1}$. For each $\tau \geq 0$, let*

$$E_\tau = \sum_{t=0}^{\tau-1} \left(V_t - \langle V_t, a_t b_t^\top \rangle a_t b_t^\top \right)$$

Then, there exists some absolute constants $C > 0, c > 0$ such that for any $\tau \in \mathbb{Z}^+$, and $0 < \delta < 1/e$, for any $\theta > 0$, with probability at least $1 - \delta$ given $\mathcal{F}_{\tau-1}$,

$$\|E_\tau\|_F \leq c \cdot \theta \sum_{t=0}^{\tau-1} dm \log^3 \left(\frac{C\tau m}{\delta} \right) \|V_t\|_F^2 + \frac{1}{\theta} \log \left(\frac{Cm^2 d \tau}{\delta} \right).$$

Proof Let $V_t = P\Sigma Q^\top = \sum_{i=1}^r \sigma_i p_i q_i^\top$ be the singular value decomposition of V_t . Extend $\{q_1, \dots, q_r\}$ to an orthonormal basis $\{q_1, \dots, q_m\}$ of $\mathbb{R}^m$. Define the events

$$A_t = \left\{ \sqrt{d}L_\delta < \|a_t\| < \sqrt{3d \log(C/\delta)} \right\}$$

and

$$B_{t,i} = \left\{ L_\delta < |b_t^\top q_i| < \sqrt{3 \log(C/\delta)} \right\}, \quad B_t = \bigcap_{i=1}^m B_{t,i}.$$

Let R be an orthogonal matrix $RR^\top = R^\top R = I$, and notice that Ra has the same distribution as a , and that A_t is the same event when $\|a_t\|$ is replaced with $\|Ra_t\|$. Therefore,

$$R\mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t] R^\top = \mathbb{E}_{\mathcal{F}_{t-1}}[(Ra_t)(Ra_t)^\top; A_t] = \mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t]$$

Now, if M is a symmetric matrix such that $RM R^\top = M$ for all R , it must be that M is a multiple of the identity. Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t] = \alpha I_d.$$

We then get

$$\begin{aligned} \alpha d &= \text{Tr}\left(\mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t]\right) = \mathbb{E}_{\mathcal{F}_{t-1}}[\|a_t\|^2; A_t] \\ &= \mathbb{E}_{\mathcal{F}_{t-1}}[(\|a_t\|^2 - d); A_t] + d\mathbb{P}(A_t) = d\mathbb{P}(A_t) \end{aligned}$$

The most important fact for this entire proof, is to notice that the events $B_{t,i}$ are independent. This is because, since $q_i \perp q_j$ for $i \neq j$, the $b_t^\top q_i$ and $b_t^\top q_j$ are independent given $\mathcal{F}_{t-1}$. Thus, if $i \neq j$,

$$\begin{aligned} \mathbb{E}_{\mathcal{F}_{t-1}}[(Q^\top b_t)(Q^\top b_t)_{i,j}^\top; B_t] &= \mathbb{E}_{\mathcal{F}_{t-1}}[(q_i^\top b_t)(q_j^\top b_t); B_t] \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k \neq i,j} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}[q_i^\top b_t; B_{t,i}] \cdot \mathbb{E}_{\mathcal{F}_{t-1}}[q_j^\top b_t; B_{t,j}] \\ &= 0 \end{aligned}$$

because if Z is $\mathcal{N}(0, 1)$ and $A \in \sigma(Z)$ is a symmetric event, we have $\mathbb{E}[Z; A] = 0$. When $i = j$,

$$\mathbb{E}_{\mathcal{F}_{t-1}}[(q_i^\top b_t)^2; B_t] = \mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{k \neq i} B_{t,k}\right) \cdot \mathbb{E}_{\mathcal{F}_{t-1}}[(q_i^\top b_t)^2; B_{t,i}] = \mathbb{P}_{\mathcal{F}_{t-1}}(B_t)$$

by our judicious choice of $B_{t,i}$ (cf. the lemma before this).

As $Q \in \mathcal{F}_{t-1}$, we thus have

$$\begin{aligned} \mathbb{E}_{\mathcal{F}_{t-1}}[Q^\top b_t b_t^\top; B_t] &= \mathbb{E}_{\mathcal{F}_{t-1}}[(Q^\top b_t)(Q^\top b_t)^\top; B_t] \cdot Q^\top \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}(B) I \cdot Q^\top = \mathbb{P}_{\mathcal{F}_{t-1}}(B) Q^\top. \end{aligned}$$

Therefore,

$$\mathbb{E}_{\mathcal{F}_{t-1}}[V_t - a_t a_t^\top V_t b_t b_t^\top; A_t, B_t] = V_t \mathbb{P}_{\mathcal{F}_{t-1}}(A_t \cap B_t) - \mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top P \Sigma Q^\top b_t b_t^\top; A_t, B_t]$$

Now,

$$\begin{aligned} \mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top P \Sigma Q^\top b_t b_t^\top; A_t, B_t] &= \mathbb{E}_{\mathcal{F}_{t-1}}[a_t a_t^\top; A_t] P \Sigma \cdot \mathbb{E}_{\mathcal{F}_{t-1}}[Q^\top b_t b_t^\top; B_t] \\ &= P \Sigma Q^\top = V_t \end{aligned}$$

Therefore, upon defining

$$Q_t = (V_t - a_t a_t^\top V_t b_t b_t^\top) \mathbb{1}_{A_t \cap B_t}$$

we have

$$\mathbb{E}_{\mathcal{F}_{t-1}}[Q_t] = 0.$$

The next part of the proof is showing that Q_t is $\text{nSG}(C\sqrt{mn}\text{polylog}(dm/\delta)\|V\|_F)$. First, notice that on B_t ,

$$\|b_t\|^2 = \sum_{i=1}^m |q_i^\top b_t| \leq 3m \log(C/\delta)$$

and recall that A_t was defined so that

$$\|a_t\| \leq \sqrt{3d \log(C/\delta)}.$$

So on $A_t \cap B_t$ we have

$$\|a_t a_t^\top V_t b_t b_t^\top\|_F = |a_t^\top V b_t| \|a_t\| \|b_t\| \leq |a_t^\top V b_t| \sqrt{9md \log^2(C/\delta)}$$

Our main goal now is to show that $a_t^\top V b_t$ is subgaussian with parameter $O(\text{polylog}(1/\delta)\|V\|_F)$. To simplify notation, we will drop the t s and conditioning, it won't be important for the argument. Recall that for $Z \sim \mathcal{N}(0, \sigma^2)$, if $t\sigma^2 < \frac{1}{2}$,

$$\mathbb{E}\left[e^{tZ^2}\right] = (1 - 2t\sigma^2)^{-1/2}$$

Then given b , since $a^\top V b$ has distribution $\mathcal{N}(0, \|Vb\|^2)$, we see immediately that whenever $t\|Vb\|^2 < \frac{1}{4}$,

$$\mathbb{E}\left[e^{t(a^\top V b)^2} \mid b\right] = (1 - 2t\|Vb\|^2)^{-1/2} \leq e^{2t\|Vb\|^2}$$

where the latter bound comes from, if $x \leq \frac{1}{4}$,

$$\log\left((1 - 2x)^{-1/2}\right) = -\frac{1}{2} \log(1 - 2x) \leq \frac{1}{2} \cdot \frac{2x}{1 - 2x} \leq 2x$$

While it is now tempting to use $\|Vb\| \leq \|V\|_F \cdot \|b\|$, this leads to a bad bound. Instead, observe that Vb is $\mathcal{N}(0, VV^\top)$ distributed.

Now recall that for $Z \sim \mathcal{N}(0, A)$, if $t < 1/(2\lambda_{\max}(A))$,

$$\mathbb{E}\left[e^{t\|Z\|^2}\right] = \det(I - 2tA)^{-1/2}$$

Therefore, as long as $t < 1/(8\sigma_{\max}(V))$, we have that

$$\begin{aligned} \mathbb{E}\left[e^{2t\|Vb\|^2}\right] &= \prod_{i=1}^r (1 - 4t\sigma_i^2)^{-1/2} = \exp\left(\sum_{i=1}^r -\frac{1}{2} \log(1 - 4t\sigma_i^2)\right) \\ &\leq \exp\left(4t \sum_{i=1}^r \sigma_i^2\right) = \exp(4t\|V\|_F^2) \end{aligned}$$

We are now almost done. Observe, on the event B ,

$$\|Vb\|^2 = b^\top V^\top Vb = \sum_{i=1}^r \sigma_i^2 b^\top q_i \leq 3 \log(C/\delta) \sum_{i=1}^r \sigma_i^2 = 3 \log(C/\delta) \|V\|_F^2$$

Now, $t\|Vb\|^2 \leq t3\|V\|_F^2 \log(C/\delta)$, so choosing $t = 1/(24\|V\|_F^2 \log(C/\delta))$, we will satisfy all the bounds above ($\sigma_{\max}(V) \leq \|V\|_F$), and get

$$\mathbb{E}\left[e^{t(a^\top Vb)^2}\right] = \mathbb{E}\left[\mathbb{E}\left[e^{t(a^\top Vb)^2} \mid b\right]\right] \leq e^{4t\|V\|_F^2} \leq \exp\left(\frac{1}{6 \log(C/\delta)}\right) \leq e^{1/6} \leq 2$$

Therefore, $a_t^\top V_t b_t$ is $\text{nSG}(\sqrt{24 \log(C/\delta)}\|V\|_F)$. Combining the previous bounds on $\|a\|$ and $\|b\|$, we get

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[\exp\left(\frac{\|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t \cap B_t}\|_F^2}{216md \log^3(C/\delta) \|V\|_F^2}\right)\right] \leq 2$$

Therefore, using $\|Q_t\|_F^2 \leq \|V_t\|^2 + \|a_t a_t^\top V_t b_t b_t^\top \mathbb{1}_{A_t \cap B_t}\|$, for some absolute constant c' ,

$$\mathbb{E}_{\mathcal{F}_{t-1}}\left[\exp\left(\frac{\|Q_t\|_F^2}{c' m d \log^3(C/\delta) \|V\|_F^2}\right)\right] \leq 2$$

This proves that Q_t is $\text{nSG}(c' \log^{3/2}(C/\delta)\|V\|_F)$.

Thus, by Lemma 1, for some absolute constant c , with probability at least $1 - \delta$,

$$\left\|\sum_{t=0}^{\tau-1} Q_t\right\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} m d \log^3(C/\delta) \|V_t\|_F^2 + \frac{1}{\theta} \log \frac{2mn}{\delta}.$$

Now, on $\bigcap_{t=0}^{\tau-1} A_t \cap B_t$,

$$\sum_{t=0}^{\tau-1} Q_t = \sum_{t=0}^{\tau-1} (V_t - a_t a_t^\top V_t b_t b_t^\top)$$

As $\mathbb{P}_{\mathcal{F}_{t-1}}(B_{t,i}) \geq 1 - \delta$, by union bound $\mathbb{P}_{\mathcal{F}_{t-1}}(B_t) \geq 1 - m\delta$. Further, recall that $\mathbb{P}_{\mathcal{F}_{t-1}}(A_t) \geq 1 - \delta$. Therefore,

$$\mathbb{P}_{\mathcal{F}_{t-1}}\left(\bigcap_{t=0}^{\tau-1} A_t \cap B_t\right) \geq 1 - \tau(m+1)\delta$$

so, with probability at least $1 - \delta - \tau(m+1)\delta$,

$$\|E_\tau\| \leq c \cdot \theta \sum_{t=0}^{\tau-1} m d \log^3(C/\delta) \|V_t\|_F^2 + \frac{1}{\theta} \log\left(\frac{2md}{\delta}\right).$$

After rescaling δ to $\delta/(1 + \tau(m+1))$, we have arrived at the desired result. ■

3.2 Sub-Weibell Random Variables

Definition 7 Random variable $X \in \mathbb{R}$ is $\text{SW}(K, \alpha)$ for some $K, \alpha > 0$ if

$$\mathbb{P}(|X| \geq s) \leq 2 \exp\left(-(s/K)^{1/\alpha}\right) \quad \forall s \geq 0$$

We will need the following lemmas.

Lemma 8 Suppose X and Y are $\text{SW}(K_X, \alpha)$ and $\text{SW}(K_Y, \alpha)$ respectively. Then XY is $\text{SW}(C(K_X \cdot K_Y), 2\alpha)$ and $X + Y$ is $\text{SW}(C(K_X + K_Y), \alpha)$ for some absolute constant C .

Lemma 9 Suppose $X_1, \dots, X_n$ are identically distributed $\text{SW}(K', \alpha)$. Then for some absolute constant $c > 0$, for all $s \geq ncK'$, we have

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq s\right) \leq \exp\left(-\left(\frac{s}{ncK'}\right)^{1/\alpha}\right).$$

We will need to frequently bound sums of $\|P_t\|^{2k}$.

Lemma 10 Suppose $a_0, \dots, a_{\tau-1} \sim \mathcal{N}(0, I_d)$, and $b_0, \dots, b_{\tau-1} \sim \mathcal{N}(0, I_m)$. Then for any $k > 0$ there exists absolute constants $c, C > 0$ such that for any $0 < \delta < 1$, w.p. $\geq 1 - \delta$,

$$\sum_{t=0}^{\tau-1} \|a_i b_i^\top\|_F^{2k} \leq \tau C c^{2k} d^k m^k (1 + \log(1/\delta))^{2k}$$

When $\delta < 1/e$,

$$\sum_{t=0}^{\tau-1} \|a_i b_i^\top\|_F^{2k} \leq 2\tau C c^{2k} d^k m^k \log(1/\delta)^{2k}.$$

Proof $(a_i)_k^2$ and $(b_j)_l^2$ are $\text{SW}(1, 1)$. Thus their product is $\text{SW}(c, 2)$. Thus $\|a_i b_i^\top\|_F^2 = \sum_{k,l} (a_i)_k^2 (b_j)_l^2$ is $\text{SW}(cmd, 2)$ for some absolute constant c . Thus, $\|a_i b_i^\top\|_F^{2k}$ is $\text{SW}(c^k m^k d^k, 2k)$. Therefore by the lemma, for $s \geq \tau C c^k m^k d^k$,

$$\mathbb{P}\left(\sum_{t=0}^{\tau-1} \|a_i b_i^\top\|_F^{2k} \geq s\right) \leq \exp\left(-\left(\frac{s}{\tau C c^k d^k m^k}\right)^{1/2k}\right)$$

the lemma follows by choosing $s = (1 + \log(1/\delta))^{2k} \tau C c^k d^k m^k$. ■

4 Function Decrease

First, we prove the following crucial lemma.

Lemma 11 (Function decrease for rank r symmetric EGGROLL) Suppose at each time t , the algorithm performs the update step with rank $r \leq dm$,

$$X_{t+1} = X_t - \eta g_u^r(X_t)$$

where

$$g_\sigma^r(X_t) := \frac{1}{r} \sum_{i=1}^r \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i}$$

where $P_{t,i} = a_{t,i} b_{t,i}^\top$ where each $a_{t,i}, b_{t,i}$ are drawn i.i.d. from $\mathcal{N}(0, I_d)$, and $\mathcal{N}(0, I_m)$ respectively, and $\sigma > 0$ is the smoothing radius.

Then there exists absolute constants $c > 0, C \geq 1$ such that, for any $T \in \mathbb{Z}^+$ and $1 \leq \tau' \leq T$, and $0 < \delta < 1/e$, then upon defining $\mathcal{H}_\tau(\delta)$ to be the event on which the inequality

$$\begin{aligned} f(X_\tau) - f(X_0) &\leq -\frac{\eta}{2} \sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 \\ &\quad + \eta^2 \frac{cdm\chi^4}{r} \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

holds, (where $\chi := \log(Cm^2 drT/\delta)$), we have

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \geq 1 - \frac{3\tau'}{T} \delta$$

Proof For each $t \in \{-1, \dots, \tau\}$, let

$$\mathcal{F}_t = \sigma(X_0, \{a_{0,i}\}_{i=1}^r, \dots, \{a_{t,i}\}_{i=1}^r, \{b_{0,i}\}_{i=1}^r, \dots, \{b_{t,i}\}_{i=1}^r)$$

be the sigma-algebra containing the information available at time t , and as usual write $P_{t,i} = a_{t,i} b_{t,i}^\top$.

Observe that, by mean value theorem, there exists some $s_{t,i} \in [-1, 1]$ such that

$$g_\sigma(X_t) = \frac{1}{r} \sum_{i=1}^r \frac{f(X_t + \sigma P_{t,i}) - f(X_t - \sigma P_{t,i})}{2\sigma} P_{t,i} = \frac{1}{r} \sum_{i=1}^r \nabla f(X_t + \sigma s_{t,i} P_{t,i}) P_{t,i}$$

Then we have that

$$X_{t+1} - X_t = -\eta \cdot \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i})$$

$$\begin{aligned}
f(X_{t+1}) - f(X_t) &\leq \langle X_{t+1} - X_t, \nabla f(X_t) \rangle + \frac{L}{2} \|X_{t+1} - X_t\|^2 \\
&\leq -\eta \cdot \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 \\
&\quad - \eta \cdot \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle \\
&\quad + \frac{L\eta^2}{2} \left\| \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2
\end{aligned}$$

Now notice, by using $ab \leq (a^2 + b^2)/2$,

$$-\frac{\eta}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle \leq \frac{\eta}{2r} \sum_{i=1}^r (|\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \sigma^2 L^2 \|P_{t,i}\|^4)$$

Also, by $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$,

$$\begin{aligned}
&\left\| \frac{1}{r} \sum_{i=1}^r (\langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} + \langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2 \\
&\leq 2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{2}{r} \sum_{i=1}^r \|\langle \nabla f(X_t + \sigma_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i}\|^2 \\
&\leq 2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{2\sigma^2 L^2}{r} \sum_{i=1}^r \|P_{t,i}\|^6
\end{aligned}$$

Therefore,

$$\begin{aligned}
f(X_{t+1}) - f(X_t) &\leq -\frac{\eta}{2r} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \frac{\eta\sigma^2 L^2}{2r} \sum_{i=1}^r \|P_{t,i}\|^4 \\
&\quad + L\eta^2 \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 + \frac{\eta^2 L^3 \sigma^2}{r} \sum_{i=1}^r \|P_{t,i}\|^6
\end{aligned}$$

Now,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 \leq 2 \left\| \frac{1}{r} \sum_{i=1}^r (\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\|^2 + 2\|\nabla f(X_t)\|^2$$

By our proposition, with probability $1 - \delta$,

$$\left\| \sum_{i=1}^r (\nabla f(X_t) - \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i}) \right\| \leq c \cdot \theta d m r \log^3 \left(\frac{Crm}{\delta} \right) \|\nabla f(X_t)\|^2 + \frac{1}{\theta} \log \left(\frac{Cm^2 dr}{\delta} \right)$$

As $\nabla f(X_t)$ is fixed given $\mathcal{F}_{t-1}$, optimizing over θ , this is at most

$$\sqrt{cdmr} \log(Cm^2 dr/\delta)^2 \|\nabla f(X_t)\|.$$

Therefore for some irrelevant absolute constant c ,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\|^2 \leq c \frac{dm}{r} \log(Cm^2 dr/\delta)^4 \|\nabla f(X_t)\|^2 \quad (1)$$

By union bound, this will hold jointly for all $0 \leq t \leq \tau' - 1$ with probability at least $1 - \tau'\delta$.

Now by old lemma, for some absolute constants C, c possibly different from before, with probability at least $1 - \delta$,

$$\frac{1}{r} \sum_{t=0}^{\tau-1} \sum_{i=1}^r \|P_{t,i}\|^{2k} \leq 2\tau C c^{2k} (dm)^k \log(1/\delta)^{2k}.$$

If k is fixed, by union bound, this will hold jointly for all $1 \leq \tau \leq \tau'$ with probability $1 - \tau'\delta$. Therefore, it will hold for both $k = 2$ and 3 , for all $1 \leq \tau \leq \tau'$, with probability at least $1 - 2\tau'\delta$.

Therefore,

$$\begin{aligned} f(X_\tau) - f(X_0) &\leq -\frac{\eta}{2r} \sum_{t=0}^{\tau-1} \sum_{i=1}^r |\langle \nabla f(X_t), P_{t,i} \rangle|^2 + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log(1/\delta)^4 \\ &\quad + \eta^2 \frac{cdm}{r} \log(Cm^2 dr/\delta)^4 \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log(1/\delta)^6 \end{aligned}$$

holds for all $1 \leq \tau \leq \tau'$ with probability at least $1 - 3\tau'\delta$. Equivalently,

$$\mathbb{P}\left(\bigcap_{\tau=1}^{\tau'} \mathcal{H}_\tau(\delta)\right) \geq 1 - 3\tau'\delta$$

Similarly, we see that, as when τ is fixed we need only Equation 6 to hold jointly over $0 \leq t \leq \tau - 1$, while the bound on the perturbation norms hold with probability at least $1 - 2\delta$, so,

$$\mathbb{P}(\mathcal{H}_\tau(\delta)) \geq 1 - (\tau + 2)\delta$$

After rescaling δ to δ/T , we have arrived at the desired result. ■

We will need the following Bernstein inequality for sub-exponential random variables (Vershynin, 2018, Theorem 2.8.1).

Lemma 12 (Bernstein concentration inequality) *Let $X_1, \dots, X_n$ be independent, mean-zero, σ -subexponential random variables, i.e.*

$$\mathbb{E}\left[\exp\left(\frac{|X_i|}{\sigma}\right)\right] \leq 2 \quad \text{for all } i \in [n].$$

Then there exists an absolute constant c such that for any $s \geq 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq s\right) \leq \exp\left(-c \min\left\{\frac{s^2}{n\sigma^2}, \frac{s}{\sigma}\right\}\right).$$

We then prove the following lemma.

Lemma 13 *There exists an absolute constant $c_2 \geq 1$ such that, if*

$$t_f(\delta) = \frac{c_2}{m} \log\left(\frac{T}{\delta}\right), \quad \delta > 0,$$

and

$$\mathcal{B}_{t_0}(\delta; k) = \bigcup_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}$$

we have

$$\mathbb{P}(\mathcal{B}_{t_0}(\delta; k)) \geq 1 - \frac{\delta}{T}$$

for $0 < \delta < 1$, $t_0 \geq 0$ and $k \geq t_f(\delta)$.

Proof By using the same technique from the proof of Theorem 3, we know that the set of random variables $\{\|\nabla f(X_t)\|^2 - |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2\}_{i=1}^m$ are independent, $c\|\nabla f(X_t)\|^2$ subexponential for some absolute constant $c > 0$ given $\mathcal{F}_{t-1}$. Now define

$$E_t = \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}$$

Therefore,

$$\begin{aligned} \mathbb{P}_{\mathcal{F}_{t-1}}(E_t) &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 \geq \|\nabla f(X_t)\|^2\right) \\ &= \mathbb{P}_{\mathcal{F}_{t-1}}\left(\sum_{i=1}^m \left(\|\nabla f(X_t)\|^2 - |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2\right) > \frac{m}{2} \|\nabla f(X_t)\|^2\right) \\ &\leq \exp\left(-c' \min\left\{\frac{m}{4c^2}, \frac{m}{2c}\right\}\right) \leq \exp(-c''m) \end{aligned}$$

for some absolute constant c'' , where the last line follows by using Bernstein concentration inequality.

Now,

$$\begin{aligned} \mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1} E_t\right) &= \mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i}^\top \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}\right) \\ &= \mathbb{E}\left[\prod_{t=t_0}^{t_0+k-2} \mathbb{1}_{E_t} \mathbb{E}_{\mathcal{F}_{t_0+k-2}}[\mathbb{1}_{E_{t_0+k-1}}]\right] \leq \cdots \leq \exp(c''mk) \end{aligned}$$

therefore by choosing $k \geq t_f(\delta)$ we get

$$\mathbb{P}\left(\bigcap_{t=t_0}^{t_0+k-1} \left\{ \frac{1}{m} \sum_{i=1}^m |a_{t,i} \nabla f(X_t) b_{t,i}|^2 < \frac{1}{2} \|\nabla f(X_t)\|^2 \right\}\right) \leq \frac{\delta}{T},$$

which completes the proof. ■

Lemma 14 *Define*

$$\mathcal{A}_t(\delta) := \left\{ \|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{8t_f(\delta)} + \eta\sigma L^2 \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \right\}$$

where $t_f(\delta)$ is defined above. Then there exists an absolute constant c_2 such that, whenever η satisfies,

$$\eta L c_2 \sqrt{\frac{dm}{r}} \log\left(\frac{CdmrT}{\delta}\right)^2 \leq \frac{1}{8t_f(\delta)}, \quad (2)$$

we have

$$\mathbb{P}(\mathcal{A}_t(\delta)) \geq 1 - \frac{\delta}{T}$$

for any $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$.

Proof By L -lipschitz of ∇f ,

$$\begin{aligned} \|\nabla f(X_{t+1}) - \nabla f(X_t)\| &\leq L \|X_{t+1} - X_t\| \\ &\leq \eta L \left\| \frac{1}{r} \sum_{i=1}^r a_{t,i} a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \\ &\quad + \eta L \left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\| \end{aligned}$$

By L -lipschitz of $\nabla f(X_t)$ and the Cauchy-Schwarz inequality,

$$\left\| \frac{1}{r} \sum_{i=1}^r \langle \nabla f(X_t + \sigma s_{t,i} P_{t,i}) - \nabla f(X_t), P_{t,i} \rangle P_{t,i} \right\| \leq L\sigma \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

Now for some absolute constant $c_2 > 0$ by Theorem 6, with probability at least $1 - \delta/T$,

$$\left\| \frac{1}{r} \sum_{i=1}^r a_{t,i} a_{t,i}^\top \nabla f(X_t) b_{t,i} b_{t,i}^\top \right\| \leq c_2 \log(CdmrT/\delta)^2 \sqrt{\frac{dm}{r}} \|\nabla f(X_t)\|$$

plugging in our condition on η , we arrive at the statement of the lemma. ■

This next very important lemma is the key that will let us lower bound

$$\sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle \nabla f(X_t), P_t \rangle|^2 \geq \tilde{\Omega} \left(\sum_{t=0}^{\tau-1} \|\nabla f(X_t)\| \right)$$

in some sense.

Denote

$$\mathcal{E}(t_1, t_2, \delta) = \bigcap_{t=t_1}^{t_1+t_2-1} \left\{ \|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \cdot \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \right\}$$

Lemma 15 *Let $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$ satisfying $T \geq 2t_f(\delta) + 1$. Consider any positive integer $t'_f \leq 2t_f(\delta)$, and any $t_0 \in \{0, \dots, T - 1 - t'_f\}$. Suppose η satisfies the condition above, then on*

$$\mathcal{E}(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{A}_t(\delta) \right)$$

we have

$$\frac{1}{2} \|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq 2 \|\nabla f(X_t)\|$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$.

Proof By plugging in the definition for $\mathcal{E}_t(\delta)$ into the definition of $\mathcal{A}_t(\delta)$, we see that on the prescribed event,

$$\|\nabla f(X_{t+1}) - \nabla f(X_t)\| \leq \frac{\|\nabla f(X_t)\|}{4t_f(\delta)}$$

therefore

$$\left(1 - \frac{1}{4t_f(\delta)}\right) \|\nabla f(X_t)\| \leq \|\nabla f(X_{t+1})\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right) \|\nabla f(X_t)\|$$

which leads to

$$\left(1 - \frac{1}{4t_f(\delta)}\right)^{t-t_0} \|\nabla f(X_0)\| \leq \|\nabla f(X_t)\| \leq \left(1 + \frac{1}{4t_f(\delta)}\right)^{t-t_0} \|\nabla f(X_0)\|.$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$. Now using $\log(1+x) \leq x$ for $x \geq 0$ and $-\log(1-x) \leq x/(1-x)$ for $0 \leq x < 1$, we know that

$$\left(1 - \frac{1}{4x}\right)^{2x} \geq e^{-2/3} \geq \frac{1}{2}, \quad \left(1 + \frac{1}{4x}\right)^{2x} \leq e^{1/2} \leq 2$$

Now recalling that $t'_f \leq 2t_f(\delta)$, we are done. ■

Lemma 16 *There exists an absolute constant $c > 0$ such that for any $t \in \mathbb{N}$, the event*

$$\mathcal{G}_t(\delta) := \left\{ \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \leq c(dm)^{3/2} \log(T/\delta)^3 \right\}$$

holds with probability at least $1 - \delta/T$ for $0 < \delta \leq 1/e$.

Proof Obvious from Lemma 10 and the Cauchy-Schwarz inequality. ■

Lemma 17 *Let $0 < \delta \leq 1/e$ and $T \in \mathbb{Z}^+$ such that $T > 2t_f(\delta) + 1$. Consider any positive integer $t'_f \leq 2t_f(\delta)$ and $t_0 \in \{0, \dots, T - 1 - t'_f\}$. Suppose η satisfies the condition above. Then on the event*

$$\mathcal{E}^c(t_0, t'_f, \delta) \cap \left(\bigcap_{t=t_0}^{t_0+t'_f-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=t_0}^{t_0+t_f-1} \mathcal{G}_t(\delta) \right)$$

we have

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

for all $t \in \{t_0, \dots, t_0 + t'_f - 1\}$.

Proof Let t' be the first time in $\{t_0, \dots, t_0 + t_f - 1\}$ such that the gradient is dominated, meaning

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2 \sigma \frac{1}{r} \sum_{i=1}^r \|P_{t',i}\|^3.$$

By definition of $\mathcal{G}_{t'}(\delta)$, we must have

$$\|\nabla f(X_{t'})\| \leq 8t_f(\delta)\eta L^2 \sigma c(dm)^{3/2} \log(T/\delta)^3$$

For $j < t'$, using the idea from a previous lemma,

$$\|\nabla f(X_j)\| \leq 2\|\nabla f(X_{t'})\| \leq 16t_f(\delta)\eta L^2 \sigma c(dm)^{3/2} \log(T/\delta)^3$$

On the other hand, for $t \in \{t', \dots, t_0 + t'_f - 1\}$, by using definition of $\mathcal{A}_t(\delta)$ and $\mathcal{G}_t(\delta)$, we get

$$\begin{aligned}
\|\nabla f(X_{t+1})\| &\leq \left(1 + \frac{1}{8t_f(\delta)}\right) \|\nabla f(X_t)\| + \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\
&= \left(1 - \frac{1}{8t_f(\delta)}\right)^{t+1-t'} \|\nabla f(X_{t'})\| \\
&\quad + \sum_{i=0}^{t-t'} \left(1 - \frac{1}{8t_f(\delta)}\right)^{t-t'-i} \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\
&\leq \left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} \|\nabla f(X_t)\| \\
&\quad + 8t_f(\delta) \left(\left(1 + \frac{1}{8t_f(\delta)}\right)^{t'_f} - 1 \right) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\
&\leq e^{1/4} 8t_f(\delta) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\
&\quad + 8t_f(\delta) (e^{1/4} - 1) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 \\
&\leq 16t_f(\delta) \eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3
\end{aligned}$$

Where we used $t'_f \leq 2t_f(\delta)$ and $(1+1/(8x))^{2x} \leq e^{1/4}$ when $x > 0$. This completes the proof. ■

We are now almost ready to prove the main theorem. But first, we need the following lemma.

Lemma 18 (Function change for large τ) *Let $0 < \delta \leq 1/e$ and let $\tau \geq t_f(\delta)$ be arbitrary. Split $\{0, \dots, \tau - 1\}$ into $K := \tau/t_f(\delta)$ intervals $J_0, \dots, J_{K-1}$ of size $t_f(\delta)$. Let I_1 denote the set of indices where the gradient always dominates the noise term in J_k , i.e.*

$$I_1 := \left\{ k : \|\nabla f(X_t)\| \geq 8t_f(\delta) \eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3 \text{ for all } t \in J_k \right\}$$

Suppose we choose η satisfying

$$\eta \leq \frac{1}{8t_f(\delta)} \min \left\{ \sqrt{\frac{r}{dm}} \cdot \frac{1}{c_2 L \log(Cdmr\tau/\delta)^2}, \frac{r}{dm} \cdot \frac{1}{8c\chi^4} \right\}$$

Then on the event

$$\mathcal{E}_\tau(\delta) := \mathcal{H}_\tau(\delta) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{G}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{\tau-1} \mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta)) \right)$$

we have the following upper bound for some absolute constants c, C ,

$$\begin{aligned}
f(X_\tau) - f(X_0) &\leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\
&\quad + \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6.
\end{aligned}$$

Furthermore,

$$\mathbb{P}(\mathcal{E}_\tau(\delta)) \geq 1 - \frac{(4\tau + 2)}{T}\delta.$$

Proof Any interval $J_k = \{t_0, \dots, t_0 + t_f - 1\}$ belongs to one of the following two cases:

(Case 1) (Gradient dominates noise): This means for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

By the Lemma 15 we have

$$\min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{4} \max_{t \in J_k} \|\nabla f(X_t)\|$$

On the event $\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))$, there exists some $t \in J_k$ such that

$$\frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 \geq \frac{1}{2} \|\nabla f(X_t)\|^2$$

Therefore, using an average,

$$\begin{aligned} \frac{1}{4} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 &\geq \frac{1}{4} \min_{t \in J_k} \|\nabla f(X_t)\| \geq \frac{1}{64} \max_{t \in J_k} \|\nabla f(X_t)\|^2 \\ &\geq \frac{1}{64t_f(\delta)} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \end{aligned}$$

By choosing

$$\frac{\eta c d m \chi^4}{r} \leq \frac{1}{64t_f(\delta)}$$

we have

$$\begin{aligned} & -\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \frac{\eta^2 c d m \chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \\ & \leq -\frac{\eta}{4} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 \leq -\frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2. \end{aligned}$$

(Case 2) (Gradient does not dominate noise): This means that there exists some $t \in J_k$ such that

$$\|\nabla f(X_t)\| \leq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3$$

Then by Lemma 17, we know that, for all $t \in J_k$,

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

Every J_k falls into one of the above two cases. Now we can complete the proof.

$$\begin{aligned} & -\frac{\eta}{2} \sum_{t=0}^{\tau-1} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t=0}^{\tau-1} \|\nabla f(X_t)\|^2 \\ &= \sum_{k \in I_1} \left(-\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \right) \\ &+ \sum_{k \in I_1^c} \left(-\frac{\eta}{2} \sum_{t \in J_k} \frac{1}{r} \sum_{i=1}^r |\langle P_{t,i}, \nabla f(X_t) \rangle|^2 + \eta^2 \frac{cdm\chi^4}{r} \sum_{t \in J_k} \|\nabla f(X_t)\|^2 \right) \\ &\leq -\sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \sum_{k \in I_1^c} t_f(\delta) \left(\eta^4 \frac{cdm\chi^4}{r} 256t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6 \right) \\ &\leq -\sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} 256t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6 \end{aligned}$$

and thus by the definition of $\mathcal{H}_\tau(\delta)$ in Lemma 11,

$$\begin{aligned} f(X_\tau) - f(X_0) &\leq -\sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + \tau \eta^4 \frac{cdm\chi^4}{r} 256t_f(\delta)^2 \sigma^2 L^4 c(dm)^3 \log(T/\delta)^6 \\ &+ \tau C c^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 \tau C c^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6. \end{aligned}$$

Furthermore,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_\tau(\delta)^c) &\leq \mathbb{P}(\mathcal{H}_\tau(\delta)^c) + \sum_{t=0}^{\tau-1} \mathbb{P}(\mathcal{A}_t(\delta)^c) + \sum_{t=0}^{\tau-1} \mathbb{P}(\mathcal{G}_t(\delta)^c) + \sum_{k=0}^{K-1} \mathbb{P}(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))^c) \\ &\leq \frac{(\tau+2)}{T} \delta + \frac{\tau}{T} \delta + \frac{\tau}{T} \delta + \frac{K}{T} \delta \leq \frac{(4\tau+2)}{T} \delta \end{aligned}$$

which completes the proof. ■

Theorem 19 (Symmetric EGGROLL Convergence) *Let $0 < \delta < 1/e$ be arbitrary. Suppose we choose σ, η and T such that*

$$\begin{aligned} \sigma^2 &\leq \min \left\{ \frac{\varepsilon^2}{1024 C c^4 L^2 (dm)^2 \log(T/\delta)^4}, \frac{\varepsilon^2}{8096 r L^3 (dm)^2 \log(T/\delta)^2} \right\} \\ \eta &\leq \frac{1}{8t_f(\delta)} \min \left\{ \sqrt{\frac{r}{dm}} \cdot \frac{1}{cL \log(Cdmr\tau/\delta)^2}, \frac{r}{dm} \cdot \frac{1}{8c\chi^4} \right\}, \\ T &\geq \max \left\{ 4, \frac{512t_f(\delta)(f(X_0) - f^*)}{\eta\varepsilon^2} \right\}. \end{aligned}$$

Then with probability at least $1 - 5\delta$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Proof Without loss of generality assume that $t_f(\delta)$ divides T and split $\{0, \dots, T-1\}$ into $K = T/t_f(\delta)$ intervals $J_0, \dots, J_{K-1}$. Let I_1 denote the set of indices k such that for every $t \in J_k$,

$$\|\nabla f(X_t)\| \geq 8t_f(\delta)\eta\sigma L^2 \frac{1}{r} \sum_{i=1}^r \|P_{t,i}\|^3.$$

Denote

$$\mathcal{E}_T(\delta) := \mathcal{H}_T(\delta) \cap \left(\bigcap_{t=0}^{T-1} \mathcal{A}_t(\delta) \right) \cap \left(\bigcap_{t=0}^{T-1} \mathcal{G}_t(\delta) \right) \cap \left(\bigcap_{k=0}^{K-1} \mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta)) \right)$$

For the rest of the proof assume we are working on $\mathcal{E}_T(\delta)$. By Lemma 18, and our choices of η and δ in the statement of the theorem, we have

$$\begin{aligned} f(X_T) - f(X_0) &\leq - \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 + T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\ &\quad + TCc^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 TCc^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

Assume that there are more than $T/4$ iterations with $\|\nabla f(X_t)\| \geq \varepsilon$. Let I_ε denote the set of indices k where there is a $t \in J_k$ with $\|\nabla f(X_t)\| \geq \varepsilon$. Thus by pigeonhole principle, $|I_\varepsilon| \geq T/4t_f(\delta)$. By our choices of parameters, it can be shown that

$$16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3 < \varepsilon$$

While by Lemma 17, if $k \in I_1^c$,

$$\|\nabla f(X_t)\| \leq 16t_f(\delta)\eta\sigma L^2 c(dm)^{3/2} \log(T/\delta)^3$$

Therefore, $I_\varepsilon \subset I_1$. Now by Lemma 15, for any $k \in I_1$,

$$\frac{1}{2} \|\nabla f(X_{kt_f(\delta)})\| \leq \|\nabla f(X_t)\| \leq 2\|\nabla f(X_{kt_f(\delta)})\|, \quad \forall t \in J_k$$

This implies that for any $k \in I_\varepsilon$, $\min_{t \in J_k} \|\nabla f(X_t)\|^2 \geq \frac{1}{16}\varepsilon^2$, and consequently,

$$- \sum_{k \in I_1} \frac{\eta}{4} \min_{t \in J_k} \|\nabla f(X_t)\|^2 \leq - \sum_{k \in I_\varepsilon} \frac{\eta}{2} \cdot \frac{\varepsilon^2}{16} \leq - \frac{T}{4t_f(\delta)} \cdot \frac{\eta}{4} \cdot \frac{\varepsilon^2}{16} = - \frac{T\eta\varepsilon^2}{256t_f(\delta)}$$

Hence,

$$\begin{aligned} f(X_T) - f(X_0) &\leq - \frac{T\eta\varepsilon^2}{256t_f(\delta)} + T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4 (dm)^3 \log(T/\delta)^6 \\ &\quad + TCc^4 \eta \sigma^2 L^2 (dm)^2 \log\left(\frac{T}{\delta}\right)^4 + \eta^2 TCc^6 \sigma^2 L^3 (dm)^3 \log\left(\frac{T}{\delta}\right)^6 \end{aligned}$$

By our choice of parameters,

$$\begin{aligned} & T\eta^4 \frac{cdm\chi^4}{r} t_f(\delta)^2 \sigma^2 L^4(dm)^3 \log(T/\delta)^6 + TCc^4 \eta \sigma^2 L^2(dm)^2 \log\left(\frac{T}{\delta}\right)^4 \\ & + \eta^2 TCc^6 \sigma^2 L^3(dm)^3 \log\left(\frac{T}{\delta}\right)^6 \leq \frac{T\eta\varepsilon^2}{512t_f(\delta)} \end{aligned}$$

We thus get

$$f(X_T) - f(X_0) \leq -\frac{T\eta\varepsilon^2}{512t_f(\delta)}$$

Therefore, as long as

$$T \geq \frac{512t_f(\delta)(f(x_0) - f^*)}{\eta\varepsilon^2}$$

we would get $f(X_T) < f^*$, a contradiction. Thus, we may conclude that on the event $\mathcal{E}_T(\delta)$ there are at most $T/4$ iterations for which $\|\nabla f(X_t)\| \geq \varepsilon$.

Finally,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_T^c) & \leq \mathbb{P}(\mathcal{H}_T^c) + \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{A}_t(\delta)) + \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{G}_t(\delta)) + \sum_{k=0}^{K-1} \mathbb{P}(\mathcal{B}_{kt_f(\delta)}(\delta; t_f(\delta))) \\ & \leq \frac{(T+2)\delta}{T} + \delta + \delta + \frac{K\delta}{T} \leq 5\delta. \end{aligned}$$

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